

Exercise (From HOTS): A coin-making machine produces pennies. Each penny is manufactured to have a probability P of turning up heads. However, the machine draws P randomly from the uniform distribution on $[0, 1]$ so P can differ for each coin produced. A coin pops out of the machine. You flip it once, and it comes up heads. Given this information, what is the (conditional) ^{cumulative} distribution function $F_{P|H}(p)$ for the probability of a head for that coin (where "H" denotes conditioning on the head)?

What is the (conditional) distribution for the probability of a head if you flip the coin 1,000 times and get 750 heads?

Assume make n flips. $X = \#$ of heads $0 \leq P \leq 1$

Find $P(P \leq p | X=x)$ for $0 \leq p \leq 1$

Note that $P(P \leq p | X=x) = \int_0^p f_{P|X}(u|x) du$

$$= \int_0^p \left[\frac{f_{X|P}(x|u) f_P(u)}{\int_{-\infty}^{\infty} f_{X|P}(x|v) f_P(v) dv} \right] du \quad \text{Using a version of Bayes Rule}$$

$$\text{Since } f_{P|X} = \frac{f_{P,X}}{f_X} = \frac{f_{X|P} f_P}{\int_{-\infty}^{\infty} f_{X|P}(x|v) f_P(v) dv}$$

Using one of LFTP

Know $f_P(p) = 1$, $0 \leq p \leq 1$

$$f_{X|P}(x|p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x=0,1,\dots,n$$

$$= \frac{\int_0^p \binom{n}{x} u^x (1-u)^{n-x} 1 \, du}{\int_0^1 \binom{n}{x} v^x (1-v)^{n-x} 1 \, dv}$$

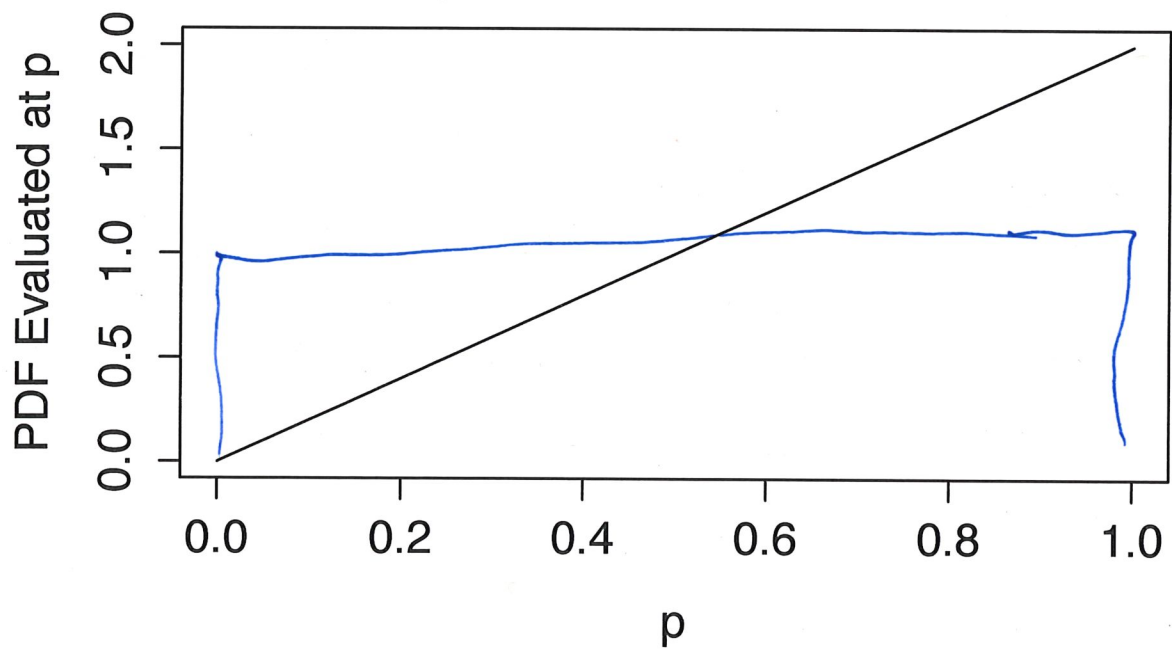
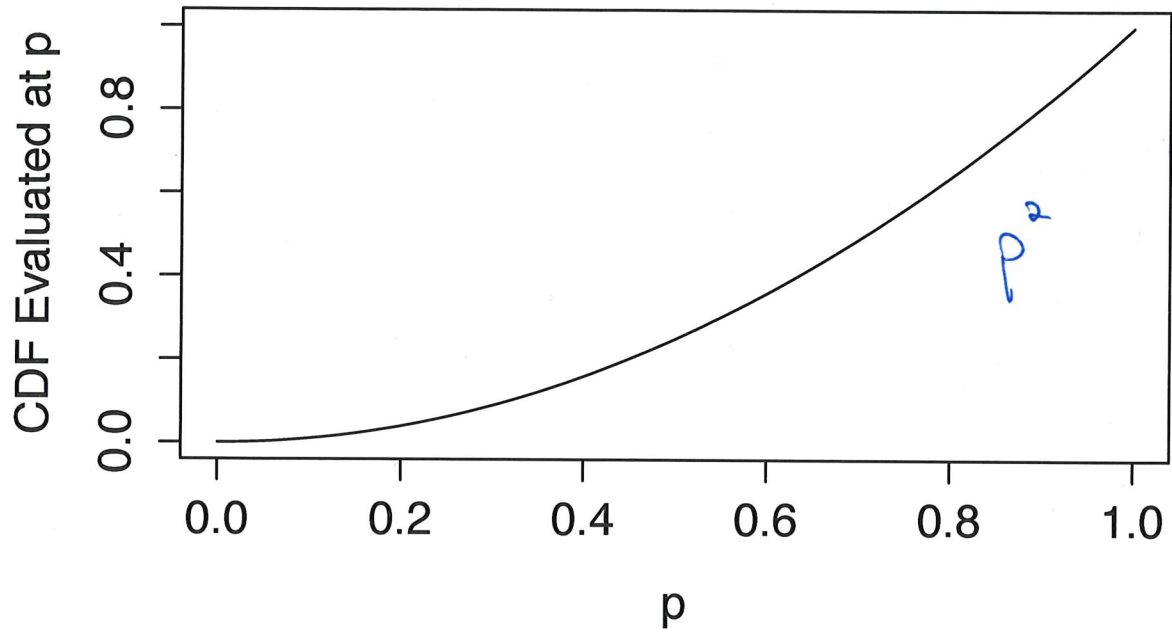
$$= \int_0^p \frac{\Gamma(n+2)}{\Gamma(x+1)\Gamma(n-x+1)} u^x (1-u)^{n-x} \, du$$

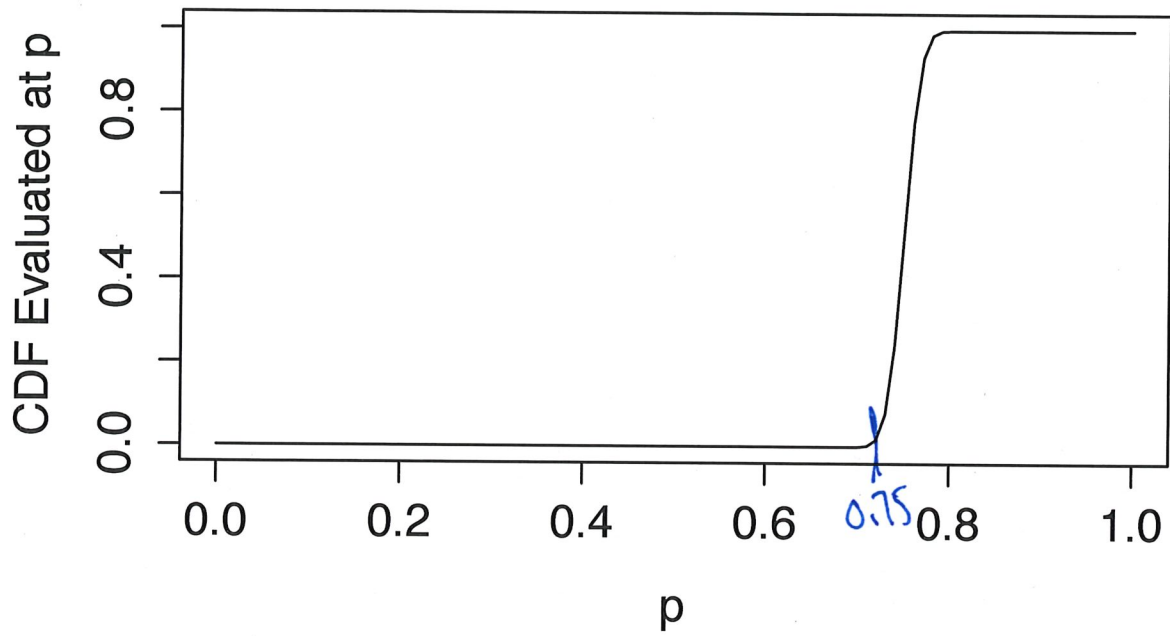
$$= P(U \leq p), \quad U \sim \text{Beta}(x+1, n-x+1)$$

Given x heads in n tosses, P has the $\text{Beta}(x+1, n-x+1)$ distribution

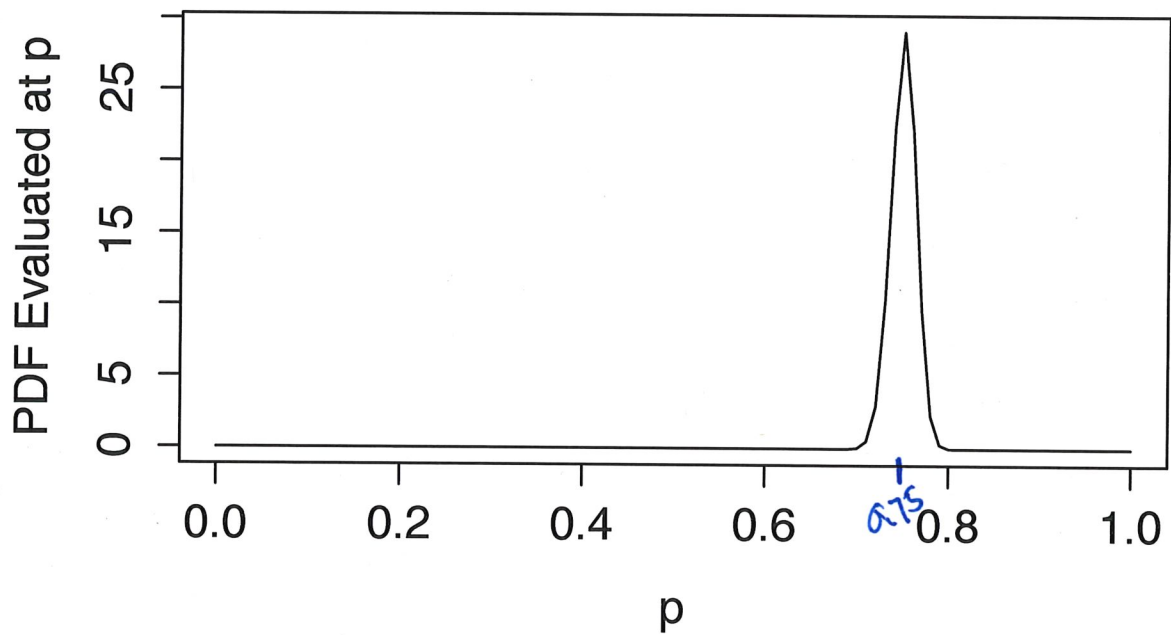
If $n=1, x=1$, $\text{Beta}(2, 1)$, i.e.

$$= \int_0^p \frac{\Gamma(3)}{\Gamma(2)\Gamma(1)} u \, du = p^2$$





Case where $n=1000$, $X=750$



Iterated Conditioning

There is convenient shorthand for the above version of the Law for Total Probability for Expectations. We often write

$$E(X) = E(\underbrace{E(X|Y)}_{\text{a RV}})$$

This expression works for both discrete and continuous random variables.

compare with $E(X) = \int_{-\infty}^{\infty} E(X|Y=y) f_Y(y) dy$

It also holds that

$$V(X) = V(E(X|Y)) + E(V(X|Y))$$

Exercise: Explain the logic behind the iterated conditioning formula.

Note that $E(X|Y)$ is a RV, i.e. is a function on Ω

Let $Z = E(X|Y)$, then $Z(\omega) = E(X|Y=Y(\omega))$ $\omega \in \Omega$

i.e. Z captures all of the possible values of Y to be conditioned on

for outcome $\omega_1 \in \Omega$, $Y = Y(\omega_1)$

" " ω_2 , $Y = Y(\omega_2)$

So, $E(Z) = E(E(X|Y))$ takes a weighted average of all possible ω (all possible $Y(\omega)$), with weighting provided by f_Y

Exercise: The lifetime of a part can be modelled using an Exponential(λ) distribution. These parts are produced in lots, and it is most realistic to assume that λ varies from lot to lot. Based on experience, the manufacturer models the lot-to-lot variability in λ using the pdf

$$f_{\Lambda}(\lambda) = 4\lambda e^{-2\lambda}, \quad \lambda > 0 \quad \Lambda \sim \text{Gamma}(2, 2)$$

Determine the unconditional probability density function for the length of life of a part. What is the probability that a randomly chosen part lasts longer than 2 time units? What is the expected length of the life of a part?

$X =$ length of life for randomly chosen part

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X|\Lambda}(x|\lambda) f_{\Lambda}(\lambda) d\lambda \\ &= \int_0^{\infty} \underbrace{\lambda e^{-\lambda x}}_{\text{exponential}} 4\lambda e^{-2\lambda} d\lambda, \quad x > 0 \end{aligned}$$

$$= \frac{8}{(x+2)^3}, \quad x > 0 \quad \text{use Math Facts}$$

$$P(X > 2) = \int_2^{\infty} \frac{8}{(x+2)^3} dx$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \dots$$

$$\text{Or : } E(X) = E(E(X | \Lambda)) \\ = E\left(\frac{1}{\Lambda}\right) \neq \frac{1}{E(\Lambda)}$$

$$= \int_0^{\infty} \left(\frac{1}{\lambda}\right) 4\lambda e^{-2\lambda} d\lambda \quad \text{Rule for Lazy Stat.}$$

$$= 2$$

Exercise: Consider a Poisson process with rate λ . Suppose that "events" are of two types: "Type I" and "Type II." An event is of Type I with probability p , and the type is determined independently from event to event. Let X denote the number of events in one time unit, ^{so that} and assume that X is $\text{Poisson}(\lambda)$. Let Y denote the number of events in one time unit that are of Type I. Find $E(Y)$ and $V(Y)$.

$$E(Y) = E(E(Y|X))$$

$$= E(Xp) \quad Y|X \sim \text{binomial}(X, p)$$

$$= pE(X) = p\lambda$$

$$V(Y) = V(E(Y|X)) + E(V(Y|X))$$

$$= V(Xp) + E(Xp(1-p))$$

$$= p^2 V(X) + p(1-p)E(X)$$

$$= \lambda p^2 + \lambda p(1-p)$$

$$= \lambda p$$

Is $Y \sim \text{Poisson}(\lambda p)$? Yes, we have not proven that yet.

Part 5: Moment Generating Functions

Suppose random variable X has some univariate distribution, discrete or continuous.

We define $E(X)$, $E(X^2)$, $E(X^3)$, $\dots$ to be the **moments of the distribution of X** . As we've already seen, $E(X^k)$ can be finite, infinite, or not exist.

The **moment generating function (MGF)** of X is

$$M_X(t) = E(e^{tX}).$$

Note that it is a function of t .

The MGF is simply the expected value of some function of X , and hence can be found using the Rule for the Lazy Statistician.

Exercise: Derive the MGF for the Poisson(λ) distribution.

$$X \sim \text{Poisson}(\lambda)$$

$$M_X(t) = E(e^{tx})$$

$$= \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{Rule for Lazy Stat.}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!}$$

$$= e^{-\lambda} e^{\lambda e^t}$$

$$= e^{\lambda(e^t - 1)}$$

$$\left(e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!} \right)$$

Exercise: Derive the MGF for the Exponential(λ) distribution.

$$M_X(t) = \frac{\lambda}{\lambda - t} \quad > t < \lambda$$

Name	Density/Mass Function	MGF	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$ $-\infty < x < \infty, \sigma > 0$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$	μ	σ^2
Log-Normal	$\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x) - \mu)^2}{2\sigma^2}\right),$ $0 < x < \infty, \sigma > 0$	—	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$
Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x},$ $0 < x < \infty, \alpha, \beta > 0$	$\left(\frac{\beta}{\beta-t}\right)^\alpha, t < \beta$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Exponential	$\beta e^{-\beta x},$ $0 < x < \infty, \beta > 0$	$\frac{\beta}{\beta-t}, t < \beta$	$\frac{1}{\beta}$	$\frac{1}{\beta^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1},$ $0 < x < 1, \alpha, \beta > 0$	$1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^k}{k!}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Uniform (continuous)	$\frac{1}{b-a},$ $a \leq x \leq b$	$\frac{e^{bt}-e^{at}}{t(b-a)}, t \neq 0$ $1, t = 0$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Bernoulli	$p^x (1-p)^{1-x},$ $x = 0, 1, 0 \leq p \leq 1$	$pe^t + (1-p)$	p	$p(1-p)$
Binomial	$\binom{n}{x} p^x (1-p)^{n-x},$ $x = 0, 1, \dots, n, 0 \leq p \leq 1$	$[pe^t + (1-p)]^n$	np	$np(1-p)$
Negative Binomial	$\binom{r+x-1}{x} p^r (1-p)^x,$ $x = 0, 1, \dots, r \in \{1, 2, \dots\}, 0 \leq p \leq 1$	$\left(\frac{p}{1-(1-p)e^t}\right)^r, t < \log(1/(1-p))$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Geometric	$(1-p)^x p,$ $x = 0, 1, \dots, 0 \leq p \leq 1$	$\frac{p}{1-(1-p)e^t}, t < \log(1/(1-p))$	$\frac{1-p}{p}$	$\frac{(1-p)}{p^2}$
Poisson	$\frac{e^{-\lambda} \lambda^x}{x!},$ $x = 0, 1, \dots, \lambda > 0$	$e^{\lambda(e^t-1)}$	λ	λ
Hypergeometric	$\frac{\binom{A}{x} \binom{B}{n-x}}{\binom{A+B}{n}},$ $x = 0, 1, \dots, n$	—	$\frac{nA}{A+B}$	$\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$

Using the MGF

1. Calculating Moments

Successive derivatives of the MGF, evaluated at $t = 0$, return the moments of the random variable:

$$E(X^k) = \left. \frac{\partial^k}{\partial t^k} M_X(t) \right|_{t=0}$$

Exercise: Why does this work? Sketch it out in the continuous case.

$$\frac{\partial^k}{\partial t^k} M_X(t) = \frac{\partial^k}{\partial t^k} \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$$

$$= \int_{-\infty}^{\infty} \frac{\partial^k}{\partial t^k} e^{tx} f_X(x) dx$$

$$= \int_{-\infty}^{\infty} x^k e^{tx} f_X(x) dx$$

plug in $t=0$

$$= \int_{-\infty}^{\infty} x^k f_X(x) dx = E(X^k)$$

Exercise: Find the moments of the Poisson(λ) distribution.

$$X \sim \text{Poisson}(\lambda)$$

$$M_X(t) = e^{\lambda(e^t - 1)}$$

$$\frac{\partial}{\partial t} M_X(t) = e^{\lambda(e^t - 1)} \lambda e^t$$

$$\text{ply in } t=0, \text{ get } E(X) = \lambda$$

2. Uniqueness of the MGF

The uniqueness result is the following: If $M_X(t) = M_Y(t)$ for all t , then X and Y have the same distribution.

It is interesting to note that two random variables can have matching moments, i.e., $E(X^k) = E(Y^k)$ for $k = 1, 2, \dots$, but X and Y have different distributions.

Exercise: Suppose that X has the Exponential(λ) distribution, and $k > 0$ is some constant. Does kX have an Exponential distribution?

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

$$\text{Let } Y = kX$$

$$M_Y(t) = E(e^{tY}) = E(e^{t(kX)}) = M_X(tk)$$

$$= \frac{\lambda}{\lambda - tk} = \frac{\lambda/k}{\lambda/k - t}, \text{ matches}$$

the Exponential(λ/k) MGF

So, Y is $\text{Exp}(\lambda/k)$

Exercise: Suppose that the number of events during an interval of length one time unit is a random variable X which has the $\text{Poisson}(\lambda)$ distribution. For each of these events, there is probability p that it is of "Type I." The determination of whether or not an event is of "Type I" is independent from event to event. Let Y denote the number of "Type I" events during the time interval. Show that Y has the Poisson distribution.

We know $X \sim \text{Poisson}(\lambda)$, $Y|X \sim \text{binomial}(X, p)$

$$M_Y(t) = E(e^{tY}) = E(E(e^{tY} | X))$$

$$= E[(pe^t + (1-p))^X] \quad \text{using the fact that } Y|X \sim \text{binomial}(X, p)$$

$$= \sum_{k=0}^{\infty} (pe^t + (1-p))^k \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{using rule for Lazy Stat.}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda(pe^t + (1-p)))^k}{k!}$$

$$= e^{-\lambda} e^{\lambda(pe^t + (1-p))}$$

$$= e^{\lambda p(e^t - 1)}, \text{ So } Y \sim \text{Poisson}(\lambda p)$$